BT-104 – BASIC ELECTRICAL & ELECTRONICS ENGINEERING

UNIT-WISE 5 MOST IMPORTANT QUESTIONS

- Based on **PYQ repetition + syllabus weightage + exam probability**
- Strictly **5 units** (no extra / no mismatch)
- RGPV-oriented (numerical + theory mix)

PYQs analysed from: **May-2019, Nov-2019, Jun-2020, Dec-2023, Jun-2022, Nov-2022, Jun-2023, Dec-2024, Jun-2025**

UNIT–I: DC CIRCUITS

👉 **Most repeated unit (numericals every year)**

Top 5 Important Questions

1️⃣ **State and explain Kirchhoff's Current Law (KCL) and Voltage Law (KVL). Solve a numerical.**

Asked in: **2019, 2022, 2025**

2️⃣ **Explain Superposition Theorem and solve a circuit using it.**

Asked in: **2019, 2020, 2023, 2025** 🔥

3️⃣ **Find current/voltage using Mesh or Nodal Analysis.**

Asked in: **2020, 2023, 2024, 2025**

4️⃣ **Explain and solve problems on Star–Delta / Delta–Star transformation.**

Asked in: **2019, 2020, 2023, 2024**

5️⃣ **Explain dependent and independent sources with neat sketches.**

Asked in: **2023, 2024**

UNIT–II: AC CIRCUITS

👉 **Guaranteed numerical + theory every year**

Top 5 Important Questions

1️⃣ **Define RMS value, average value, form factor and peak factor of AC quantity.**

Asked in: **2020, 2022, 2023, 2024, 2025** 🔥

2️⃣ **Solve numericals on RLC series circuit (Impedance, current, power factor).**

Asked in: **2022, 2023, 2024, 2025**

3️⃣ **Explain active, reactive and apparent power in AC circuits.**

Asked in: **2019, 2020, 2022, 2023**

4️⃣ **Three-phase balanced star/delta system – line & phase values with phasor diagram.**

Asked in: **2020, 2022, 2024** 🔥

5️⃣ **Explain resonance in RLC circuit (series/parallel).**

Asked in: **2019, 2023**

UNIT–III: MAGNETIC CIRCUITS & TRANSFORMER

👉 **Transformer = fixed long question**

Top 5 Important Questions

1️⃣ **Explain construction, working principle and EMF equation of single-phase transformer.**

Asked in: **2019, 2020, 2023, 2025** 🔥

2️⃣ **OC & SC test on transformer and determination of equivalent circuit parameters.**

Asked in: **2020, 2023, 2024**

3️⃣ **Losses in transformer and efficiency calculation numericals.**

Asked in: **2019, 2022, 2025**

4️⃣ **Explain magnetization characteristics of ferromagnetic materials (B-H curve).**

Asked in: **2020, 2023, 2024**

5️⃣ **Define self-inductance & mutual inductance. Derive their relation.**

Asked in: **2022, 2023**

UNIT–IV: ELECTRICAL MACHINES

👉 Mostly **theory + diagrams**

Top 5 Important Questions

1️⃣ **Explain construction and working principle of DC machine with diagram.**

Asked in: **2019, 2024, 2025**

2️⃣ **Explain principle of operation of 3-phase induction motor.**

Asked in: **2020, 2022, 2023, 2025** 🔥

3️⃣ **Explain torque–slip characteristics of induction motor.**

Asked in: **2023, 2024**

4️⃣ **Explain losses occurring in electrical machines.**

Asked in: **2019, 2023, 2024**

5️⃣ **Numericals on slip, synchronous speed, rotor speed.**

Asked in: **2023, 2024**

UNIT–V: BASIC ELECTRONICS

👉 **Short notes + diagrams compulsory**

Top 5 Important Questions

1️⃣ **Draw and explain V-I characteristics of PN junction diode.**

Asked in: **2020, 2023, 2025** 🔥

2️⃣ **Explain working of BJT and its configurations (CE/CB/CC).**

Asked in: **2019, 2022, 2023, 2024**

3️⃣ **Explain Half adder and Full adder with truth table.**

Asked in: **2020, 2022, 2024**

4️⃣ **Explain JK flip-flop / RS flip-flop with logic diagram.**

Asked in: **2019, 2020, 2023, 2024**

5️⃣ **Explain De-Morgan's Theorem with example.**

Asked in: **2019, 2020, 2022, 2024**

EXAM STRATEGY (VERY IMPORTANT)

Unit	Importance
Unit I (DC Circuits)	⭐⭐⭐⭐⭐
Unit II (AC Circuits)	⭐⭐⭐⭐⭐
Unit III (Transformer)	⭐⭐⭐⭐
Unit IV (Machines)	⭐⭐⭐
Unit V (Electronics)	⭐⭐⭐

👉 **Units I + II = 50% paper numericals**

👉 **Transformer + Electronics = easy theory marks**

FULL PREDICTED MODEL PAPER (2025)

Time: 3 Hours

Maximum Marks: 70

Instructions

1. Attempt **any FIVE** questions.
2. All questions carry **equal marks (14 marks each)**.
3. Draw neat diagrams wherever required.
4. Assume suitable data if required.

Q1. (UNIT–I: DC CIRCUITS)

a) State and explain Kirchhoff's Laws.

Apply KVL to find the current in a given DC circuit. (7)

b) Explain the Superposition Theorem.

Solve a simple circuit using superposition method. (7)

Q2. (UNIT–I: NETWORK THEOREMS)

a) Explain Thevenin's Theorem with necessary steps. (7)

b) Find the Thevenin equivalent circuit across given terminals and calculate load current. (7)

Q3. (UNIT–II: AC CIRCUITS)

a) Define RMS value, average value, form factor, and peak factor of AC quantity. (7)

b) An RLC series circuit is connected to AC supply.

Find **impedance, current, power factor and power consumed**. (7)

Q4. (UNIT–II: THREE-PHASE AC CIRCUITS)

a) Explain balanced three-phase system with neat phasor diagram. (7)

b) Derive the relation between line and phase quantities for

i) Star connection

ii) Delta connection (7)

Q5. (UNIT–III: TRANSFORMER)

a) Explain construction and working principle of a single-phase transformer with diagram. (7)

b) Derive the EMF equation of a transformer. (7)

Q6. (UNIT–III: TRANSFORMER PERFORMANCE)

a) Explain losses in transformer and methods to improve efficiency. (7)

b) A transformer has given losses.

Calculate **efficiency at full load**. (7)

Q7. (UNIT–IV: ELECTRICAL MACHINES)

a) Explain construction and working principle of DC machine with diagram. (7)

b) Explain working principle of three-phase induction motor.

Draw **torque–slip characteristics**. (7)

Q8. (UNIT–V: BASIC ELECTRONICS)

a) Draw and explain V-I characteristics of PN junction diode. (7)

b) Write short notes on any TWO: (7)

- BJT and its configurations (CE, CB, CC)
- Half adder and Full adder
- JK flip-flop
- De-Morgan’s Theorem

- Applications of induction motor